

Sector-neutral carry in a synthetic 24-asset market

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1. Hypothesis

Assets with a persistently high `carry_score` earn a small, steady premium over assets with a low one, and the premium is best captured within each sector so that sector and market moves cancel. In real markets carry is the return for holding: a coupon, a dividend, a funding gap. This market is synthetic and the mechanism is not visible, but the designers labelled the column carry, made it slow-moving, and a slow characteristic is the natural place for a premium to sit. I treated the label as a hint and let the data decide.

2. Evidence

I scanned every provided column on development data only (Jan 2021 to Jun 2024). Each day I ranked the 24 assets by the column and by the return they would earn if bought at the next open, and correlated the two rankings (rank IC). Three columns had $|t| > 2$ with the same sign every year; everything else was noise, despite names like sentiment and flow.

Signal	Direction	1-day IC (t)	5-day IC (t)	Names switching side per day
return_1d	reversal	-0.048 (-6.5)	-0.010 (-0.6)	12.4 of 24
carry_score	buy high	+0.024 (+3.4)	+0.044 (+2.9)	0.9 of 24
return_5d	reversal	-0.018 (-2.6)	+0.002 (0.1)	5.3 of 24

Sorted into four carry buckets, next-day return rose monotonically: -1.7, +1.5, +2.6, +4.1 bps/day. Top minus bottom earned 5.8 bps/day ($t = 3.1$) and was positive on only 52% of days: a small, consistent edge, not a large one. One-day reversal was the strongest signal but flips half the book daily; carry barely trades. Conflicting evidence: the carry edge was strong in 2023 and weaker in 2021–22 and the first half of 2024, so it was not uniform.

3. Construction

Every fifth session, rank the four assets in each sector by `carry_score`. Long the top two, short the bottom two. Size each name by the inverse of its 20-day realized volatility, then scale so every sector is dollar-neutral, the whole book is dollar-neutral, and gross exposure is 100%. Hold the weights between rebalances. The function is stateless and re-derives the rebalance date from the history it receives. Parameters were fixed before any backtest: 5-session cadence to limit churn, gross 1.0 as a conservative default well inside the 200% limit, two names per side because a sector only has four assets. Nothing was fitted.

4. Risk

Maximum single-asset weight is about 6%, far inside the 20% limit. Sector nets are zero by construction, so no sector or market bet can decide the outcome. Development annual volatility was 3.9% and the worst drawdown 3.2%. All six sectors contributed positively over development; the largest single-asset contribution was 3.9% of NAV. Expected failure: the carry premium fades or reverses, in which case the book earns roughly zero minus costs. Zero cash yield and no borrow cost are simplifications; a short book of 50% would carry real financing cost.

5. Costs

Costs use the rules' spread-plus-impact formula from the previous close. Turnover was 14.5x per year, mostly small rebalancing trades as volatilities drift. Total development cost 3.2% of NAV over 3.5 years, about 0.9% a year.

Cost multiplier	1.0x	1.5x	2.0x
Development Sharpe	1.34	1.23	1.11
Total costs	3.2%	4.7%	6.3%

Costs shaped two decisions. First, one-day reversal was rejected: on development it loses in every window after costs (Sharpe -2.6, turnover 271x, costs 55% of NAV over the period), even at 60% gross. Second, I tested down-weighting wide-spread names (divide by sqrt of spread): costs fell 15% but Sharpe fell from 1.34 to 1.17, so it was rejected.

6. Validation

Window	Sharpe	Total return	Max DD	Positive months
Dev tune 2021–2022	0.89	+7.4%	-3.2%	
Dev hold-out 2023–Jun 2024	2.00	+11.8%	-2.5%	
Development, all	1.34	+20.2%	-3.2%	29 of 42
Validation Jul–Dec 2024, frozen	-1.75	-3.6%	-5.7%	2 of 6

The strategy was frozen (git tag frozen-before-validation) and validation was run once. It lost. One diagnostic pass afterwards, with no change to the strategy, showed why: the carry rank IC by period was +0.013 (2021–22), +0.047 (2023), +0.024 (H1 2024) and -0.008 in validation. The signal itself faded to zero; costs were 0.4% and construction held (five of six sectors lost, no single name dominated). Sensitivity on development: rebalance every 1/3/5/10/20 sessions gave Sharpe 1.09/1.17/1.34/1.28/1.24; gross 0.6/1.0/1.4 gave 1.35/1.34/1.34; one name per side instead of two gave 1.14. No setting flipped the sign. No changes were made after validation.

7. Limitations

The edge is small (about 5 bps/day gross) and regime-dependent: it was carried by 2023 and had already weakened before validation. The scan's 16 signals times two horizons is a mild multiple-testing risk, mitigated by requiring the same sign in every year and a monotonic bucket pattern. The mechanism behind carry_score is unknown, so there is no economic reason to expect it to return in 2025. Reversal signals strengthened in validation (return_1d IC -0.11) but remain untradable at these cost levels. If 2025 resembles late 2024, this strategy will lose a few percent; if it resembles 2023, it will earn 5–10% at low volatility.

8. Next step

Test a simple on/off switch: hold the carry book only when the trailing 60-day carry IC is positive, otherwise sit in cash. It is one line of logic, it would have cut exposure in late 2024, and it can be evaluated on development without touching validation again.